Tutorial Sheet No. 4

Matrices and Linear Algebra

Ka Hoi Conrad Ho

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1 Introduction: Scalars, Vectors, and Tensors

What is a Tensor?

Mathematics organises data by how many indices are needed to specify a value:

Name	Rank	Example
Scalar	0	Temperature: $T = 25$
Vector	1	Velocity: (v_x, v_y, v_z)
Matrix	2	Transformation: A_{ij}
Higher tensor	≥ 3	Stress field: T_{ijk}

A **scalar** needs zero indices; a **vector** needs one; a **matrix** needs two. Each rank adds one dimension of information. In this sheet we focus on rank-1 (vectors) and rank-2 (matrices) objects.

Everyday Analogy

A single temperature reading is a scalar. A list of temperatures for seven days is a vector (one index: which day). A table of hourly temperatures for each day is a matrix (two indices: which hour, which day). Adding more factors — say, location — builds higher-rank tensors.

1) State whether each quantity below is a scalar, vector, or matrix. Justify your answer briefly.

   a) The mass of an object.

   b) The displacement of a particle in three dimensions.

   c) The exchange rates between five currencies, recorded as a 5×5 table.

   d) The speed of light, $c \approx 3 \times 10^8$ m s^{-1}.

   e) A list of three exam marks.

2) A spreadsheet records the monthly sales (in units) of four products over twelve months.

   a) What is the shape (dimensions) of this data as a matrix?

   b) What does one row represent? What does one column represent?

   c) How many individual numbers (entries) are stored in total?

3) *Concept question.* A photograph is stored as a grid of pixels; each pixel has three colour values (R, G, B). Explain why the full image data is a rank-3 tensor rather than a matrix.

2 Notation and Representing Data

Vectors and Matrix Notation

Column vector (written vertically, shape $n \times 1$):

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

Row vector (written horizontally, shape $1 \times n$):

$$\mathbf{u} = \begin{pmatrix} u_1 & u_2 & u_3 \end{pmatrix}$$

Matrix (shape $m \times n$, m rows, n columns):

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

The entry in row i, column j is denoted a_{ij}.

Representing a System of Equations as a Matrix

The system

$$\begin{cases} 2x + 3y = 8 \\ 5x - y = 2 \end{cases}$$

is written in matrix form $A\mathbf{x} = \mathbf{b}$ as:

$$\underbrace{\begin{pmatrix} 2 & 3 \\ 5 & -1 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} x \\ y \end{pmatrix}}_{\mathbf{x}} = \underbrace{\begin{pmatrix} 8 \\ 2 \end{pmatrix}}_{\mathbf{b}}$$

1) Write each of the following as a column vector:

   a) The point $(4, -1, 7)$ in 3-D space.

   b) The coefficients of $3x^2 - x + 5$ (from highest to lowest degree).

2) Identify the shape of each matrix and state a_{23} where applicable:

$$P = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}, \quad Q = \begin{pmatrix} 7 \\ 0 \\ -3 \end{pmatrix}, \quad R = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

3) A student scores 78, 85, 92, 67 in four subjects. Write this as:

   a) a row vector,

   b) a column vector.

4) Write each system in matrix form $A\mathbf{x} = \mathbf{b}$. State the matrix A, the vector $\mathbf{x}$, and the vector $\mathbf{b}$.

   a) $\begin{cases} 3x + y = 7 \\ x - 2y = 1 \end{cases}$

   b) $\begin{cases} x + 2y - z = 4 \\ 2x - y + 3z = 0 \\ -x + y + z = 5 \end{cases}$

5) The augmented matrix of a system is:

$$\left(\begin{array}{cc|c} 4 & -1 & 9 \\ 2 & 3 & 7 \end{array}\right)$$

   Write out the corresponding system of equations.

3 Matrix Addition and Scalar Multiplication

Rules

Two matrices can be added only if they have the **same shape**.
Addition: Add entry by entry.

$$(A+B)_{ij} = a_{ij} + b_{ij}$$

Scalar multiplication: Multiply every entry by the scalar k.

$$(kA)_{ij} = k\,a_{ij}$$

Key properties:

- $A + B = B + A$ (commutative)
- $(A + B) + C = A + (B + C)$ (associative)
- $k(A + B) = kA + kB$ (distributive)

Common Mistake

You **cannot** add a 2×3 matrix to a 3×2 matrix, even though they contain the same number of entries. Shape must match exactly.

Let

$$A = \begin{pmatrix} 3 & -1 \\ 2 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 4 \\ -2 & 5 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 1 & -1 \\ 2 & 3 & 4 \end{pmatrix}$$

1) Compute: $A + B$, $A - B$, $3A$, $2A - 4B$.

2) Can $A + C$ be computed? Explain.

3) Find the matrix X such that $A + X = B$.

4) Given

$$M = \begin{pmatrix} k & 3 \\ -1 & 2 \end{pmatrix}, \quad N = \begin{pmatrix} 4 & k \\ 3k & 0 \end{pmatrix}$$

find the value of k such that the entry $(M + N)_{11} = 9$.

5) Two shops record their stock:

$$\text{Shop A} = \begin{pmatrix} 50 & 30 \\ 20 & 45 \end{pmatrix}, \quad \text{Shop B} = \begin{pmatrix} 25 & 60 \\ 35 & 15 \end{pmatrix}$$

where rows are product types and columns are sizes. Find the total stock matrix. What does each entry represent?

4 Matrix Multiplication

Definition

For A of shape $m \times n$ and B of shape $n \times p$, the product $C = AB$ has shape $m \times p$, with:

$$c_{ij} = \sum_{k=1}^{n} a_{ik}\, b_{kj}$$

In words: entry (i, j) of C is the **dot product of row i of A and column j of B.**

Matrix Multiplication is NOT Commutative

In general, $AB \neq BA$. Always check that the inner dimensions match before multiplying. For a product $A_{m\times n} \cdot B_{n\times p}$: the inner dimensions ($n$ and n) must agree.
Aside: For some matrix A and B, we can see if they commute by calculating what is *commutator* $[A, B]$ where it is defined:

$$[A, B] = AB - BA$$

If A and B commutes, then $[A, B] = 0$.
If A and B does not commute, then $[A, B] \neq 0$.

Worked Example

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 1 \cdot 5 + 2 \cdot 6 \\ 3 \cdot 5 + 4 \cdot 6 \end{pmatrix} = \begin{pmatrix} 17 \\ 39 \end{pmatrix}$$

Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & 3 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

1) State the shape of each possible product among A, B, C (where defined). For any undefined product, state why.

2) Compute: AB, BA. Are they equal?

3) Compute $A\mathbf{v}$ and interpret the result geometrically as a transformation of the vector $\mathbf{v}$.

4) Compute $A^2 = A \cdot A$ and $B^2 = B \cdot B$.

5) Find all 2×2 matrices X satisfying $AX = I$, where $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is the identity matrix.

6) A shop sells three products. The **quantity matrix** gives units sold in two weeks:

$$Q = \begin{pmatrix} 10 & 15 \\ 8 & 12 \\ 20 & 5 \end{pmatrix}$$

The **price vector** gives the price of each product (in \$):

$$\mathbf{p} = \begin{pmatrix} 3 \\ 5 \\ 2 \end{pmatrix}$$

a) Can you compute $Q\mathbf{p}$? What shape is the result?

b) Compute the product and explain what each entry means.

c) Which week had higher total revenue?

5 The Determinant

Determinant of a 2×2 Matrix

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$:

$$\det(A) = |A| = ad - bc$$

Geometric meaning: $|\det(A)|$ is the factor by which A scales areas. If $\det(A) = 0$, the matrix is **singular** (non-invertible) and the transformation collapses space to a lower dimension.

Determinant of a 3×3 Matrix (Cofactor Expansion)

$$\det(A) = a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Expand along any row or column; the alternating sign pattern is:

$$\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

Key Properties of Determinants

- $\det(AB) = \det(A)\det(B)$
- $\det(A^{-1}) = 1/\det(A)$
- $\det(kA) = k^n \det(A)$ for an $n \times n$ matrix
- Swapping two rows *negates* the determinant
- A row of zeros $\Rightarrow \det = 0$

1) Evaluate the determinant of each matrix:

$$P = \begin{pmatrix} 4 & 3 \\ 2 & 1 \end{pmatrix}, \quad Q = \begin{pmatrix} 5 & -2 \\ 10 & -4 \end{pmatrix}, \quad R = \begin{pmatrix} 7 & 0 \\ 0 & 3 \end{pmatrix}$$

2) Without computing, explain why $\det(Q) = 0$ for the matrix Q above.

3) Evaluate:

$$\det \begin{pmatrix} 2 & -1 & 3 \\ 0 & 4 & -2 \\ 1 & 0 & 5 \end{pmatrix}$$

Show all steps of your cofactor expansion.

4) Find all values of k for which the matrix

$$M = \begin{pmatrix} k & 2 \\ 3 & k-1 \end{pmatrix}$$

is singular.

5) If $\det(A) = 3$ and $\det(B) = -2$, find:

a) $\det(AB)$

b) $\det(A^2)$

c) $\det(3A)$ (for A a 2×2 matrix)

6) *Geometry connection.* Vectors $\mathbf{u} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ form a parallelogram. Show that the area of the parallelogram equals $\left|\det \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}\right|$.

6 The Rotation Matrix

Rotation by Angle θ (Counter-Clockwise)

To rotate a vector $\mathbf{v}$ by angle θ counter-clockwise about the origin:

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

The rotated vector is $\mathbf{v}' = R(\theta)\,\mathbf{v}$.

Important Properties

- $\det(R(\theta)) = \cos^2\theta + \sin^2\theta = 1$ (rotation preserves area)
- $R(\theta)^{-1} = R(-\theta) = R(\theta)^\top$ (undo by rotating back)
- $R(\alpha)\,R(\beta) = R(\alpha+\beta)$ (composing rotations adds angles)

Worked Example

Rotate $\mathbf{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ by $\theta = 90$:

$$R(90)\,\mathbf{v} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

As expected, the unit vector along x maps to the unit vector along y.

1) Write out explicitly the rotation matrices $R(30)$, $R(45)$, $R(180)$. Simplify all trigonometric values.

2) Apply $R(90)$ to each vector and sketch both the original and rotated vectors:

   a) $\mathbf{a} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$

   b) $\mathbf{b} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

   c) $\mathbf{c} = \begin{pmatrix} 0 \\ -3 \end{pmatrix}$

3) Verify that $R(45)\,R(45) = R(90)$ by matrix multiplication.

4) Show that $R(\theta)\,R(-\theta) = I$ for any angle θ.

5) A vector $\mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ is first rotated by 60, then by a further 30.

   a) Compute the two-step rotation as a single matrix.

   b) Compute $R(90)\,\mathbf{v}$ directly and verify you get the same result.

6) *Challenge.* A graphics engine applies $R(1)$ to a point 360 times. What is the net effect? Verify using the composition property.

7 Mixed Matrix Operations

The Transpose

The **transpose** $A^\top$ of a matrix A is obtained by swapping rows and columns: $(A^\top)_{ij} = a_{ji}$.

$$\text{If } A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}_{2\times 3}, \quad \text{then } A^\top = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}_{3\times 2}$$

Useful identities: $(AB)^\top = B^\top A^\top$, $\quad (A^\top)^\top = A$.

The Inverse of a 2×2 Matrix

If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\det(A) \neq 0$:

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Note the swap of the diagonal and the sign change of the off-diagonal entries.

Let

$$A = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -2 \\ 0 & 3 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

1) Compute $A^\top$, $B^\top$, and verify $(AB)^\top = B^\top A^\top$.

2) Find A^{-1} and B^{-1}. Verify each by checking $AA^{-1} = I$.

3) Compute $A^{-1}B$ and BA^{-1}. Are they the same? Should they be?

4) Evaluate each expression in the correct order:

   a) $A(B\mathbf{u})$

   b) $(AB)\mathbf{u}$

   c) $A\mathbf{u} + 2\mathbf{w}$ (note: $\mathbf{w}$ is a 2×1 vector; check compatibility)

   d) $\mathbf{u}^\top A\mathbf{w}$ (this is a scalar — why?)

5) Solve the matrix equation $AX = B$ for X.

6) Given $(2A - 3B)^\top = C$, find the matrix C explicitly.

7) *Error-spotting.* A student writes:

$$(A + B)^{-1} \stackrel{?}{=} A^{-1} + B^{-1}$$

Disprove this with a counterexample using A and B given above.

8 Cramer's Rule

Cramer's Rule for $n \times n$ Systems

For the system $A\mathbf{x} = \mathbf{b}$ where $\det(A) \neq 0$:

$$x_i = \frac{\det(A_i)}{\det(A)}$$

where A_i is the matrix A with its i-th column replaced by $\mathbf{b}$.

Worked Example: 2×2 System

Solve:

$$\begin{cases} 2x + y = 5 \\ x - 3y = -4 \end{cases}$$

$$A = \begin{pmatrix} 2 & 1 \\ 1 & -3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 5 \\ -4 \end{pmatrix}, \quad \det(A) = -6 - 1 = -7$$

$$x = \frac{\det\begin{pmatrix} 5 & 1 \\ -4 & -3 \end{pmatrix}}{-7} = \frac{-15 + 4}{-7} = \frac{-11}{-7} = \frac{11}{7}$$

$$y = \frac{\det\begin{pmatrix} 2 & 5 \\ 1 & -4 \end{pmatrix}}{-7} = \frac{-8 - 5}{-7} = \frac{-13}{-7} = \frac{13}{7}$$

When to Use Cramer's Rule

Cramer's rule is most elegant for 2×2 and 3×3 systems. For large systems, Gaussian elimination is more efficient. The key advantage of Cramer's rule is that it gives an explicit formula for each variable independently.

1) Use Cramer's rule to solve each system:

a) $\begin{cases} 3x - y = 7 \\ x + 2y = 4 \end{cases}$

b) $\begin{cases} -2x + 5y = 1 \\ 4x - 3y = 9 \end{cases}$

c) $\begin{cases} x + y + z = 6 \\ 2x - y + z = 3 \\ x + 2y - z = 4 \end{cases}$

2) For the system

$$\begin{cases} kx + 2y = 4 \\ 3x + ky = 6 \end{cases}$$

find the value(s) of k for which Cramer's rule fails (i.e. the system has no unique solution). What happens geometrically in each case?

3) A student uses Cramer's rule on a system and obtains $\det(A) = 0$. Without solving, what can you conclude about the system?

4) *Verification.* Solve question 1(a) above also by substitution. Confirm that both methods give the same answer.

5) Two numbers x and y satisfy: their sum is 15 and three times the first number minus the second equals 5. Set up and solve the system using Cramer's rule.

9 Dot Product and Cross Product

Dot Product (Scalar Product)

For $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$:

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3 = |\mathbf{a}||\mathbf{b}| \cos\theta$$

where θ is the angle between the vectors.

Key facts:

- Result is a **scalar**.
- $\mathbf{a} \cdot \mathbf{b} = 0 \Leftrightarrow \mathbf{a} \perp \mathbf{b}$ (for non-zero vectors).
- $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$.

Cross Product (Vector Product)

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{pmatrix}$$

Key facts:

- Result is a **vector** perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.
- $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin\theta$.
- $\mathbf{a} \times \mathbf{b} = -\,\mathbf{b} \times \mathbf{a}$ (anti-commutative).
- $\mathbf{a} \times \mathbf{b} = \mathbf{0} \Leftrightarrow \mathbf{a} \parallel \mathbf{b}$.

Let $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix}$, $\mathbf{c} = \begin{pmatrix} 0 \\ 3 \\ -1 \end{pmatrix}$.

1) Compute $\mathbf{a} \cdot \mathbf{b}$, $\mathbf{a} \cdot \mathbf{c}$, and $\mathbf{b} \cdot \mathbf{c}$.

2) Find $|\mathbf{a}|$, $|\mathbf{b}|$, and the angle between $\mathbf{a}$ and $\mathbf{b}$ (give your answer in degrees, to 1 d.p.).

3) Find a vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ using the cross product.

4) Compute $|\mathbf{a} \times \mathbf{b}|$ and use it to find the area of the parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$.

5) Verify that $\mathbf{a} \times \mathbf{b} = -\,\mathbf{b} \times \mathbf{a}$ by computing both products.

6) Determine whether $\mathbf{b}$ and $\begin{pmatrix} 8 \\ -2 \\ 4 \end{pmatrix}$ are parallel.

7) *Connection to matrices.* Express $\mathbf{a}\cdot\mathbf{b}$ as a matrix product using a row vector and a column vector. What shape is the result?

8) *Application.* A force $\mathbf{F} = \begin{pmatrix} 0 \\ 0 \\ 10 \end{pmatrix}$ N acts at a point $\mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix}$ m from a pivot. Compute the torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$ and state its direction.

10 Summary and Formulae

Quick Reference: Matrices

Shapes and compatibility:

- Addition $A_{m\times n} + B_{m\times n}$: shapes must match.
- Multiplication $A_{m\times n} \cdot B_{n\times p}$: inner dimensions must agree; result is $m \times p$.

2×2 **formulae:**

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \det(A) = ad - bc, \quad A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Rotation matrix:

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \quad \det R(\theta) = 1, \quad R^{-1}(\theta) = R(-\theta)$$

Cramer's rule:

$$A\mathbf{x} = \mathbf{b}, \quad x_i = \frac{\det(A_i)}{\det(A)}, \quad \det(A) \neq 0$$

3×3 **determinant (cofactor expansion along row 1):**

$$\det(A) = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$$

where M_{ij} is the (2×2) minor determinant obtained by deleting row i and column j.

Quick Reference: Vectors

Dot product:

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 = |\mathbf{a}||\mathbf{b}|\cos\theta$$

$\mathbf{a} \perp \mathbf{b} \Leftrightarrow \mathbf{a} \cdot \mathbf{b} = 0$.

Cross product:

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}, \quad |\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}|\sin\theta$$

Area of parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$: $|\mathbf{a} \times \mathbf{b}|$.

Angle between vectors:

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

Self-Assessment Checklist

Before your next session, make sure you can:

- ☐ Classify a quantity as scalar, vector, or matrix and state its shape.
- ☐ Write a system of equations as $A\mathbf{x} = \mathbf{b}$ and read it back out.
- ☐ Add matrices and multiply by scalars without errors.
- ☐ Multiply two matrices, checking inner dimensions first.
- ☐ Evaluate a 2×2 and 3×3 determinant.
- ☐ Find the inverse of a 2×2 matrix and verify it.
- ☐ Apply a rotation matrix to a vector.
- ☐ Solve a 2×2 or 3×3 system using Cramer's rule.
- ☐ Compute the dot product and find the angle between two vectors.
- ☐ Compute the cross product and find the area of a parallelogram.